TrES-5b

R-1871

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_210829` · created 2026-08-03T10:43:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459455.694 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.03 +/- 0.053
Transit depth (Rp/Rs)^2	0.091 +/- 0.32 [%]
Orbital Inclination (inc)	80.16 +/- 2.89
Airmass coefficient 1 (a1)	0.027 +/- 0.014
Airmass coefficient 2 (a2)	2.91 +/- 1.34
Scatter in the residuals of the lightcurve fit is	48.06 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	4.75
Transit Duration (day)	0.023 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.03 ± 0.053 vs 0.142 (deviation 2.1σ)
Duration fit vs published	0.023 d vs 0.0758 d (deviation 2.02σ)
Mid-time O–C	-1.1 min
Depth SNR	0.6
Residual scatter	48.06 %

Light curve

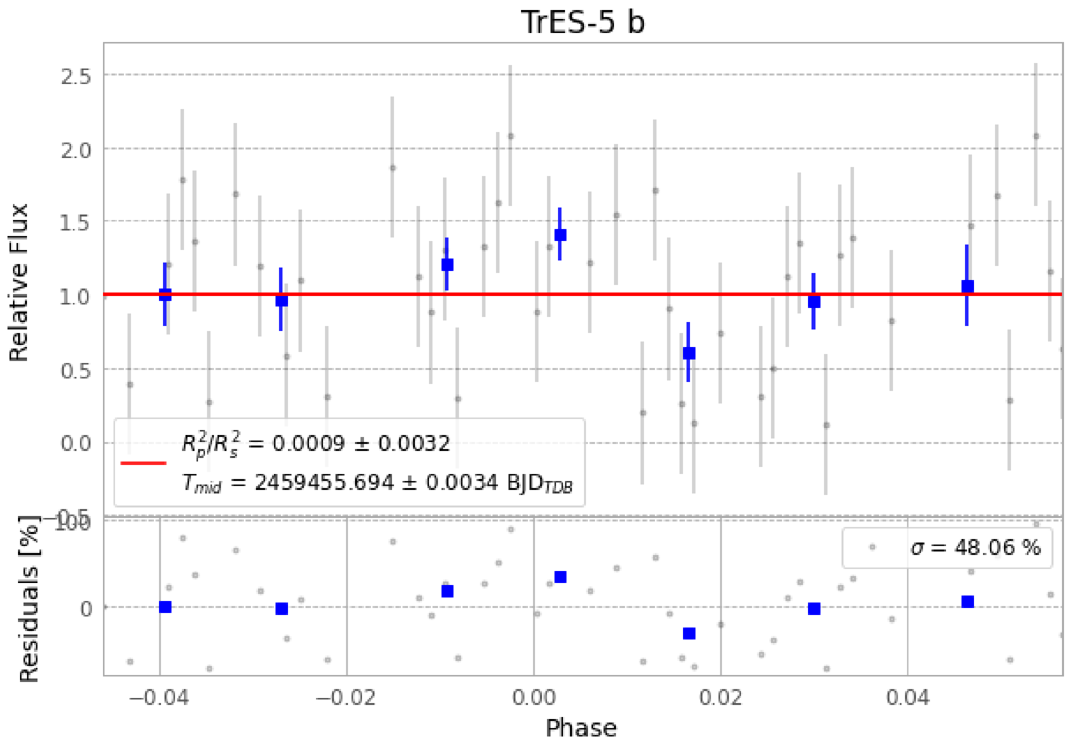

Figure 1 — FinalLightCurve_TrES-5 b_2021-08-28.png (SHA-256 eee04281bbcd3159…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [622, 180], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 732edfd373056b11…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

76 FITS frames (50 MB), `TRES-5210829025112.FITS` → `TRES-5210829063612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-210829.log (81791861c2db…), FinalLightCurve_TrES-5 b_2021-08-28.png (eee04281bbcd…), FinalLightCurve_TrES-5 b_2021-08-28.csv (b81843f4b2fe…)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)